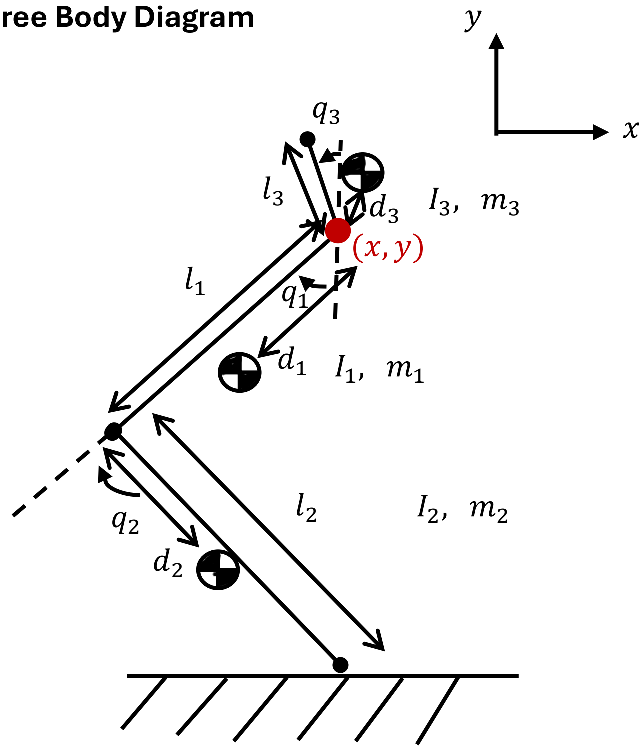
Free Body Diagram



Link 1: Hip to Knee

l_1 length of link 1, meters

I_1 moment of inertia of link 1, meters^4

m_1 mass of link 1, kg

q_1 relative angle of link 1, °

d_1 distance to Link 1 COM from torso

Link 2: Knee to Foot

l_2 length of link 2, meters

I_2 moment of inertia of link 2, meters^4

m_2 mass of link 2, kg

d_2 distance to Link 2 COM from knee

q_2 relative angle of link 2, °

Link 3: Hip to Torso

l_3 length of link 3, meters

I_3 moment of inertia of link 3, meters^4

m_3 mass of link 3, kg

d_3 distance to Link 3 COM from knee

q_3 relative angle of link 3, °

Position of Knee (Forward Kinematics)

$$p_{knee} = \begin{bmatrix} x_{knee} \\ y_{knee} \end{bmatrix} = \begin{bmatrix} x - l_1 \sin(q_1) \\ y - l_1 \cos(q_1) \end{bmatrix}$$

Position of Foot (Forward Kinematics)

$$p_{foot} = \begin{bmatrix} x_{foot} \\ y_{foot} \end{bmatrix} = \begin{bmatrix} x - \ l_1 \sin(q_1) + l_2 \sin(q_2 - q_1) \\ y - \ l_1 \cos(q_1) - l_2 \cos(q_2 - q_1) \end{bmatrix}$$

Position of Torso (Forward Kinematics)

$$p_{torso} = \begin{bmatrix} x_{torso} \\ y_{torso} \end{bmatrix} = \begin{bmatrix} x - \ l_3 \sin(q_3) \\ y + l_3 \cos(q_3) \end{bmatrix}$$

Position of COM of Link 1 (Hip to Knee)

$$p_1 = \begin{bmatrix} x_{COM1} \\ y_{COM1} \end{bmatrix} = \begin{bmatrix} x - d_1 \sin(q_1) \\ y - d_1 \cos(q_1) \end{bmatrix}$$

Position of COM of Link 2 (Knee to Foot)

$$p_2 = \begin{bmatrix} x_{COM2} \\ y_{COM2} \end{bmatrix} = \begin{bmatrix} x - \ l_1 \sin(q_1) + d_2 \sin(q_2 - q_1) \\ y - \ l_1 \cos(q_1) - d_2 \cos(q_2 - q_1) \end{bmatrix}$$

Position of COM of Link 3 (Hip to Torso)

$$p_3 = \begin{bmatrix} x_{COM3} \\ y_{COM3} \end{bmatrix} = \begin{bmatrix} x - \ d_3 \sin(q_3) \\ y + d_3 \cos(q_3) \end{bmatrix}$$

Total COM

$$p_{COM} = \begin{bmatrix} x_{COM} \\ y_{COM} \end{bmatrix} = \begin{bmatrix} \dfrac{m_1 x_{COM1} + m_2 x_{COM2} + m_3 x_{COM3}}{m_1 + m_2 + m_3} \\ \dfrac{m_1 y_{COM1} + m_2 y_{COM2} + m_3 y_{COM3}}{m_1 + m_2 + m_3} \end{bmatrix}$$

Velocity of Link 1 and Link 2

$$v_1 = \dot{p}_1 = \begin{bmatrix} \dot{x} - d_1 \cos(q_1)\, \dot{q}_1 \\ \dot{y} + d_1 \sin(q_1)\, \dot{q}_1 \end{bmatrix}$$

$$v_2 = \dot{p}_2 = \begin{bmatrix} \dot{x} - \ l_1 \cos(q_1)\, \dot{q}_1 - d_2 \cos(q_2 - q_1)\,(\dot{q}_2 - \dot{q}_1) \\ \dot{y} + \ l_1 \sin(q_1)\, \dot{q}_1 + d_2 \sin(q_2 - q_1)\,(\dot{q}_2 - \dot{q}_1) \end{bmatrix}$$

$$v_3 = \dot{p}_3 = \begin{bmatrix} \dot{x} - d_3 \cos(q_3)\, \dot{q}_3 \\ \dot{y} - d_3 \sin(q_3)\, \dot{q}_3 \end{bmatrix}$$

Kinetic Energy (T)

$$\mathrm{T}_{\mathrm{Total}} = \mathrm{T}_1 + \mathrm{T}_2 + \mathrm{T}_3$$

$$\mathrm{T}_1 \ = \frac{1}{2} m_1 v_1^T v_1 + \frac{1}{2} I_{com,1} \dot{\omega}_1^2, \qquad \mathrm{T}_2 \ = \frac{1}{2} m_2 v_2^T v_2 + \frac{1}{2} I_{com,2} {\dot{\omega}_2}^2, \qquad \mathrm{T}_3 \ = \frac{1}{2} m_3 v_3^T v_3 + \frac{1}{2} I_{com,3} {\dot{\omega}_3}^2$$

$$\dot{\omega}_1 = -\dot{q}_1, \qquad \dot{\omega}_2 = -(\dot{q}_1 - \dot{q}_2), \qquad \dot{\omega}_3 = \dot{q}_3$$

Potential Energy (U)

$$\mathrm{U}_{Total} = \mathrm{U}_1 + \mathrm{U}_2 + \mathrm{U}_3$$

$$\mathrm{U}_1 = m_1 g p_1^T \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathrm{U}_2 = m_2 g p_2^T \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathrm{U}_3 = m_3 g p_3^T \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(General) Equations of Motion

$$D(q)\ddot{q} \ + \ C(q, \dot{q})\dot{q} \ + \ G(q) = B(q)\, u + J(q)\lambda$$

where

$$B(q) = \begin{bmatrix} 0 \\ 0 \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \Gamma = \left(\frac{\partial q_1}{\partial q} \right)^T \tau$$

$$\gamma(q) = J_{st}(q)^T F_{st} \equiv 0$$

$D(q)$ – Mass Matrix, $C(q,\dot{q})$ – Coriolis Matrix, $G(q)$ – Gravity Vector, $B(q)$ – Input Mapping Matrix, $J(q)\lambda$ – constraint

<u>We will do an un-pinned model</u>

<u>(During takeoff, impact) Equations</u>

$$\begin{bmatrix} D(q) & -J_{st}(q)^T \\ J_{st}(q) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ F_{st} \end{bmatrix} = \begin{bmatrix} B(q)u - C(q,\dot{q})\dot{q} - G(q) \\ -\dot{J}_{st}(q,\dot{q})\dot{q} \end{bmatrix}$$

<u>(During Flight, no contact) Equations</u> <u>$(\gamma(q) = 0)$</u>

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = B(q)\,u$$

<u>(During touchdown, impact) Equations</u>

$$D(q)\dot{q}^+ - D(q)\dot{q}^- = J_{st}^T(q)F_{imp}$$

$$\Rightarrow \begin{bmatrix} J_{st}(q) & 0 \\ D(q) & -J_{st}^T(q) \end{bmatrix} \begin{bmatrix} \dot{q}^+ \\ F_{imp} \end{bmatrix} = \begin{bmatrix} 0 \\ D(q)\dot{q}^- \end{bmatrix}$$

<u>(During Stance, no slip, rigid contact) Equations</u>

$$\begin{bmatrix} D(q) & -J_{st}(q)^T \\ J_{st}(q) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ F_{st} \end{bmatrix} = \begin{bmatrix} B(q)u - C(q,\dot{q})\dot{q} - G(q) \\ -\dot{J}_{st}(q,\dot{q})\dot{q} \end{bmatrix}$$

Virtual Constraint

We want to enforce the holonomic virtual constraint that the COM x-direction is above the position of the foot, i.e.,

$$
\begin{aligned}
y = h(q) &= h_0(q) - h_d\big(\theta(q)\big) \\
&= p_{COM}\begin{bmatrix}1\\0\end{bmatrix} - p_{foot}\begin{bmatrix}1\\0\end{bmatrix} \\
&= 0
\end{aligned}
$$

We want to find $u(t)$ such that,

1. If $y(0) = 0$ then $y(t) = 0$ for $t \geq 0$ (invariance)
2. If $y(0) \neq 0$ then $y(t) \rightarrow 0$ as $t \rightarrow \infty$ (asymptotic attractivity)

We know the dynamics equation is,

$$D(q)\ddot{q} \; + \; C(q,\dot{q})\dot{q} \; + \; G(q) = B(q)\,u + J(q)\lambda$$

So,

$$\dot{x} = \begin{bmatrix}\dot{q}\\ \ddot{q}\end{bmatrix} = \begin{bmatrix}\dot{q}\\ D^{-1}\{-C\dot{q} - G\}\end{bmatrix} + \begin{bmatrix}0\\ D^{-1}B\end{bmatrix}u + \begin{bmatrix}0\\ D^{-1}J\end{bmatrix}\lambda$$

Let,

$$f(x) = \begin{bmatrix}\dot{q}\\ D^{-1}\{-C\dot{q} - G\}\end{bmatrix}, \qquad g_1(x) = \begin{bmatrix}0\\ D^{-1}B\end{bmatrix}, \qquad g_2(x) = \begin{bmatrix}0\\ D^{-1}J\end{bmatrix}$$

Taking the first-time derivative of $h(q)$,

$$
\begin{aligned}
\dot{y} &= \frac{dh}{dx}\frac{dx}{dt} \\
&= \frac{dh}{dx}\dot{x} \\
&= \begin{bmatrix}\frac{\partial h}{\partial q} & \frac{\partial h}{\partial \dot{q}}\end{bmatrix}\left\{\begin{bmatrix}\dot{q}\\ D^{-1}\{-C\dot{q} - G\}\end{bmatrix} + \begin{bmatrix}0\\ D^{-1}B\end{bmatrix}u + \begin{bmatrix}0\\ D^{-1}J\end{bmatrix}\lambda\right\}, \qquad \frac{\partial h}{\partial \dot{q}} = 0 \\
&= \begin{bmatrix}\frac{\partial h}{\partial q} & 0\end{bmatrix}\left\{\begin{bmatrix}\dot{q}\\ D^{-1}\{-C\dot{q} - G\}\end{bmatrix} + \begin{bmatrix}0\\ D^{-1}B\end{bmatrix}u + \begin{bmatrix}0\\ D^{-1}J\end{bmatrix}\lambda\right\} \\
\dot{y} &= \frac{\partial h}{\partial q}\dot{q}
\end{aligned}
$$

Therefore,

$$L_f h = \frac{\partial h}{\partial q}\dot{q}, \qquad L_g h = 0$$

Taking the second time derivative of $h(q)$,

$$\ddot{y} = \frac{d}{dt}\big(\dot{h}\big)$$

$$
\begin{aligned}
&= \frac{d}{dt}\left(\frac{\partial h}{\partial q}\dot{q}\right) \\
&= \frac{dL_f h}{dt} \\
&= \frac{dL_f h}{dx}\frac{dx}{dt} \\
&= \frac{dL_f h}{dx}\dot{x} \\
&= \begin{bmatrix} \frac{\partial L_f h}{\partial q} & \frac{\partial L_f h}{\partial \dot{q}} \end{bmatrix} \left\{ \begin{bmatrix} \dot{q} \\ D^{-1}\{-C\dot{q} - G\} \end{bmatrix} + \begin{bmatrix} 0 \\ D^{-1}B \end{bmatrix} u + \begin{bmatrix} 0 \\ D^{-1}J \end{bmatrix} \lambda \right\} \\
&= \begin{bmatrix} \frac{\partial}{\partial q}\left(\frac{\partial h}{\partial q}\dot{q}\right) & \frac{\partial}{\partial \dot{q}}\left(\frac{\partial h}{\partial q}\dot{q}\right) \end{bmatrix} \left\{ \begin{bmatrix} \dot{q} \\ D^{-1}\{-C\dot{q} - G\} \end{bmatrix} + \begin{bmatrix} 0 \\ D^{-1}B \end{bmatrix} u + \begin{bmatrix} 0 \\ D^{-1}J \end{bmatrix} \lambda \right\} \\
&= \begin{bmatrix} \frac{\partial}{\partial q}\left(\frac{\partial h}{\partial q}\dot{q}\right) & \frac{\partial h}{\partial q} \end{bmatrix} \left\{ \begin{bmatrix} \dot{q} \\ D^{-1}\{-C\dot{q} - G\} \end{bmatrix} + \begin{bmatrix} 0 \\ D^{-1}B \end{bmatrix} u + \begin{bmatrix} 0 \\ D^{-1}J \end{bmatrix} \lambda \right\}
\end{aligned}
$$

Since,

$$
f(x) = \begin{bmatrix} \dot{q} \\ D^{-1}\{-C\dot{q} - G\} \end{bmatrix}, \qquad g_1(x) = \begin{bmatrix} 0 \\ D^{-1}B \end{bmatrix}, \qquad g_2(x) = \begin{bmatrix} 0 \\ D^{-1}J \end{bmatrix}
$$

Using the Lie-Derivative Notation we can say,

$$
\begin{aligned}
\ddot{y} &= \frac{d}{dx}L_f h\, f(x) + \frac{d}{dx}L_f h\, g_1(x)\, u + \frac{d}{dx}L_f h\, g_2(x)\, \lambda \\
\ddot{y} &= L_f^2 h + L_{g_1}L_f h\, u + L_{g_2}L_f h\, \lambda
\end{aligned}
$$

Assuming $L_{g_1}L_f h \neq 0$, $L_2 L_f h \neq 0$, these are decoupling matrices.

Finally, we want, $y(t) \to 0$ as $t \to \infty$, and so, this is possible if we implement a PD controller such that $\ddot{y} = v$ where,

$$
v = -\left(K_p y + K_d \dot{y}\right), \qquad K_p > 0 \text{ and } K_d > 0
$$

Now, we need to determine the three unknowns, $\ddot{q},\ u,\ \lambda$, and three equations,

$$
\begin{cases}
D(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = B(q)\,u + J_{st}(q)\lambda \\
\ddot{y} = L_f^2 h + L_{g_1}L_f h\, u + L_{g_2}L_f h\, \lambda = v \\
J_{st}\ddot{q} + \dot{J}_{st}\dot{q} = 0
\end{cases}
$$

which we can solve simultaneously,

$$
\begin{bmatrix} D & -B & -J_{st} \\ 0 & L_{g_1}L_f h & L_{g_2}L_f h \\ J_{st} & 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ u \\ \lambda \end{bmatrix} = \begin{bmatrix} -C\dot{q} - G \\ v - L_f^2 h \\ -\dot{J}_{st}\dot{q} \end{bmatrix}
$$

$$\begin{bmatrix} D & -B & -J_{st} \\ 0 & L_{g_1}L_f h & L_{g_2}L_f h \\ J_{st} & 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ u \\ \lambda \end{bmatrix} = \begin{bmatrix} -C\dot{q} - G \\ \nu - L_f^2 h \\ -\dot{J}_{st}\dot{q} \end{bmatrix}$$

$$\begin{bmatrix} D & -B & -J_{st} \\ 0 & L_{g_1}L_f h & L_{g_2}L_f h \\ J_{st} & 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ u \\ \lambda \end{bmatrix} = \begin{bmatrix} -C\dot{q} - G \\ \nu - L_f^2 h \\ -\dot{J}_{st}\dot{q} \end{bmatrix}$$

Generalized Coordinates

$$(q, \dot{q}) = \left\{ \begin{bmatrix} x \\ y \\ q_1 \\ q_2 \\ q_3 \end{bmatrix}, \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} \right\}$$

Need to assume/choose numeric parameters $l_1, l_2, l_3, m_1, m_2, m_3, I_1, I_2, I_3, d_1, d_2, d_3$.

Need to determine the path of motion $q(t),\ \dot{q}(t)$

We want to find:

- Torque before take-off
- Torque during take-off
- Torque while in the air
- Torque on landing

Path of Motion